Name: ______________________

Worksheet

Astrophysicist

Task A: Being an astrophysicist

Astrophysics bodies
Astronomers celestial
narrowed physical
observational techniques

1) Fill in the gaps.

________ is a branch of astronomy.

_______ study the universe using _______ _______ to assess and categorise various _______ bodies.

Astrophysicists have a more _______ focus. They study the physics of the universe such as how the celestial _______ interact or other _______ processes.

2) a) The table below shows the approximate average distances from Earth to some of the other planets in our solar system. Write the distances in standard form.

From	To	Distance (km)	Standard form
Earth	Mercury	91 691 000	
Earth	Venus	41 400 000	
Earth	Mars	78 340 000	
Earth	Jupiter	628 730 000	
Earth	Saturn	1 275 000 000	

2) b) In 2021, the fastest speed recorded by a spacecraft was 163 km/s. Using that speed and the formula

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Calculate how long it would take, in days, to travel from Earth to each planet.

Round your answer appropriately to the nearest day.

To	Distance (km)	Time (days)
Mercury	91 691 000	
Venus	41 400 000	
Mars	78 340 000	
Jupiter	628 730 000	
Saturn	1 275 000 000	

Task B: Determining terrain features

1) Calculate the value (to 1 d.p.) of the unknown measurement in each case.

a) $\theta° = 19°$, $a = 56$ cm, $b =$

b) $\theta° = 56°$, $a = 19$ cm, $b =$

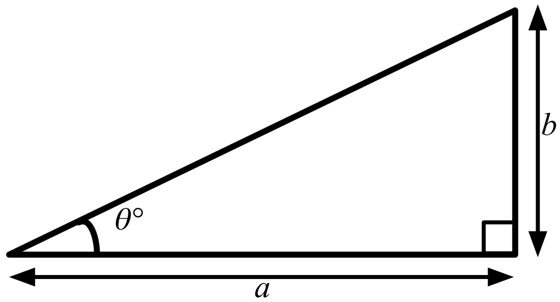
c) $\theta° = 19°$, $b = 56$ cm, $a =$

d) $\theta° = 56°$, $b = 19$ cm, $a =$

e) $a = 56$ cm, $b = 19$ cm, $\theta° =$

f) $a = 19$ cm, $b = 56$ cm, $\theta° =$

Not drawn accurately



2) The wall of a crater on Mars is 1.2 km tall. When the wall casts a shadow with length 557 m, what angle (to 1 d.p.) is the sun at?

3) Another crater on Mars has a wall which casts a shadow of length 21 m. This occurs when the sun is at an angle of 67°. What is the height (to 2 d.p.) of this wall?

4) The largest crater on the near side of the Moon is called Clavius. The walls of this crater are 6.6 km high.

Calculate three pairs of values for the angle of the sun and the corresponding length of the shadow cast by the crater wall.

E.g. When the angle of the sun is 42°, the length of the shadow is recorded as 7300 m.

When the angle of the sun is 87°, the length of the shadow is recorded as 346 m.

5) In order to calculate an estimate for the height of a crater wall, we have produced a model/sketch of the situation.

State an assumption that we have made that may affect the accuracy of our estimated value.